Student ID: 112006234

I have done this assignment without the use of any AI agent or tool

Week 1 Report

Practice Attempts on Simple R Commands in Class

```
> numbers <- c(12, 14, 14, 25, 33, 35, 38, 38, 41, 43, 45, 50,
58, 59)
> numbers[0]
numeric(0)
> numbers[1]
[1] 12
> numbers[14]
[1] 59
> numbers[15]
[1] NA
> seq(0,7)
[1] 0 1 2 3 4 5 6 7
> seq(7,0)
[1] 7 6 5 4 3 2 1 0
> seq(-1,-5)
[1] -1 -2 -3 -4 -5
> help(sum)
> sum(numbers)
[1] 505
```

Code 0.1. Creating a vector of data

```
> customers <- read.table(file = "customers.txt", header = TRUE)
> customers$age
  [1] 49 69 41 73 45 71 50 43 70 32 47 77 64 50 50 45 49 47 62 50 47 72 47
63 21 49 50 48 35 77 48 48 50 47 29
 [36] 42 42 85 45 49 45 43 49 68 42 48 72 79 48 50 47 45 30 76 31 49 74 72
48 49 73 50 47 47 83 72 75 50 50 49
 [71] 48 45 49 49 49 72 50 75 74 72 74 76 49 50 76 36 45 35 24 45 50 42 22
60 59 45 51 46 47 34 63 71 37 21 43
[106] 32 47 35 70 26 63 54 45 47 26 35 22 31 70 51 37 41 53 34 45 34 43 50
29 49 46 44 26 49 48 26 34 25 38 25
...
[386] 41 46 49 49 23 74 53 23 51 71 50 50 67 74
> ages <- customers$age
> help(length)
> length(ages)
[1] 399
```

Code 0.2. Loading data into a data frame

```
> ages <- customers$age
> hist(ages)
> plot(density(ages))
> boxplot(ages, horizontal = TRUE)
> stripchart(ages, method = "stack", add = TRUE)
> 0.1 * 10 == 1.0
[1] TRUE
> 0.1 * 10 == 3.0
[1] FALSE
> pick_ages <- ages[c(3, 12, 21)]
> sorted_ages <- sort(ages)
> sorted_ages[c(3, 12, 21)]
[1] 19 20 21
> mean(sorted_ages)
[1] 46.80702
> 0.1 == 0.10
[1] TRUE
> length(ages) / sum(ages)
[1] 0.02136432
> (0.1 * 5) - 0.5
[1] 0
> (0.1 * 3) - 0.3
[1] 5.551115e-17
```

Code 0.3. Statistical Expressions, Visualizing Distributions, and Limits of Computation

Assignment

Setting up

```
# load data file (header=TRUE means the first row has column names)
customers <- read.table("customers.txt", header = TRUE)

# extract age column as a vector out of the data frame to with it directly
ages <- customers$age
```

Code 0.4. Setting Up for the Assignment Questions

Q1. Finding the 5th element in the original list of ages.

```
ages[5]
# [5] selects the 5th position in the vector 'ages'
[1] 45
```

Code 1. Question 1

Q2. Finding the fifth lowest age.

```
sort(ages)[5]
# sort() puts values in increasing order, then [5] selects the 5th
smallest
[1] 19
```

Code 2. Question 2

Q3. Extract the five lowest ages together.

```
sort(ages)[1:5]
# 1:5 creates a sequence for the first five sorted values
[1] 18 19 19 19 19
```

Code 3. Question 3

Q4. Getting the five highest ages by sorting in decreasing order.

```
sort(ages, decreasing = TRUE)[1:5]
# decreasing=TRUE sorts from largest to smallest, then we take the first
five
[1] 85 83 82 82 81
```

Code 4. Question 4

Q5. Finding the average age.

```
mean(ages)
# mean() computes the average
[1] 46.80702
```

Code 5. Question 5

Q6. Finding the standard deviation of ages.

```
sd(ages)
# sd() calculates the standard deviation (how spread out the ages are
around the mean)
[1] 16.3698
```

Code 6. Question 6

Q7. Creating a new variable ‘age_diff’ that calculates the difference between each age and the mean age.

```
age_diff <- ages - mean(ages)
# subtracting a single value from a vector calculates the difference of
every element
```

Code 7. Question 7

Q8. Finding the average of ‘age_diff’ (as defined in Q7).

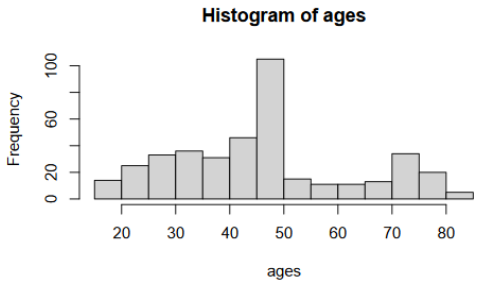
This checks whether the mean was calculated correctly and whether the differences were defined properly. In Q7, if you subtract the mean from each element of ‘ages’, the positive and negative differences should balance out.

In other words, If Q7 is correct, Q8's result must be 0 or extremely close to 0.

```
mean(age_diff)
# The mean of deviations from the mean is 0 by theory, but due to
floating-point rounding in computers, we get a value extremely close to 0
instead
[1] -1.623275e-15
```

Code 8. Question 8

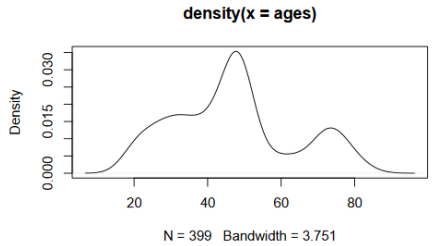
Q9. (a) Visualize the raw data as a histogram.




```
hist(ages)
# hist() groups data to show the distribution of values as a histogram
```

Code 9a. Question 9(a)

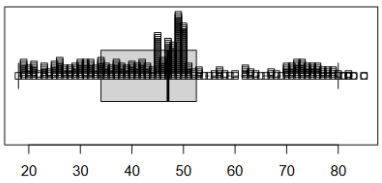
(b) Visualize the raw data as a density plot.




```
plot(density(ages))
# density() estimates a smooth distribution curve from the data
```

Code 9b. Question 9(b)

(c) Visualize the raw data as a boxplot+stripchart.




```r
boxplot(ages, horizontal = TRUE)
# boxplot() summarizes the data using quartiles and median in a horizontal
layout

stripchart(ages, method = "stack", add = TRUE)
# stripchart() adds individual data points, stacked to avoid overlapping
```

Code 9c. Question 9(c)

TSING HUA UNIVERSITY